REPORT 1: PROJECT DESCRIPTION REPORT **Project Title:** Online Library Management System (OLMS) **Institution:** Kigali Independent University (ULK)[cite: 1] **Course / Assignment:** UML Class Diagram Modelling Assignment --- ### 1. Introduction and Business Idea The **Online Library Management System (OLMS)** is an integrated software information system designed to automate and streamline the day-to-day operations of an academic or public library. Traditional library systems often rely on manual record-keeping, which can lead to inefficiencies, human errors in tracking book loans, misplacement of records, and difficulties in monitoring overdue fines. The proposed system modernizes these operations by providing a centralized digital platform to manage books, library members, staff, and borrowing transactions seamlessly. ### 2. Problem Statement and Purpose * **Problem Statement:** Manual library management leads to several administrative bottlenecks, including slow book check-out and check-in processes, inaccurate tracking of return due dates, lack of real-time visibility into book availability, and manual calculation of penalties which frequently results in discrepancies and delays. * **Purpose of the System:** The primary objective of the OLMS is to create an efficient, error-free, and automated environment. It empowers members to easily search and reserve books while providing librarians with powerful tools to manage inventory, enforce borrowing rules, track active loans, and automatically assess fines for overdue materials. ### 3. Project Scope To ensure a focused and well-defined system architecture, the project scope is categorized into included and excluded features: * **Included in Scope (Functions Covered):** * **User Management:** Managing authentication, roles, and profiles for both members and librarians. * **Book Inventory Management:** Adding, updating, removing, and searching for books within the library database. * **Borrowing Transactions:** Issuing books to members, tracking issue dates and due dates, and recording returns. * **Fine Management:** Automatically calculating overdue days and generating penalty fines for late returns. * **Excluded from Scope (Functions Not Covered):** * Direct third-party online payment gateway integration (fines are recorded and managed administratively within the system). * Built-in digital e-reader capabilities or direct reading of PDF e-books. ### 4. Main Classes and Their Responsibilities * **User (Abstract Class):** Acts as the base superclass containing common attributes (such as user ID, name, email, and phone) and general authentication methods (`login()` and `logout()`). * **Member:** Inherits from `User` and represents registered library patrons who can search for books, view their borrowing history, and reserve titles. * **Librarian:** Inherits from `User` and represents administrative staff responsible for managing the book catalog, handling physical inventory, and processing loan records. * **Book:** Represents physical or digital library assets, tracking details such as title, author, ISBN, and real-time availability status (`isAvailable`). * **BorrowRecord:** Represents the association between a member and a borrowed book, tracking issue dates, due dates, and actual return dates. * **Fine:** Manages financial penalties incurred when a `BorrowRecord` is returned past its due date. ### 5. Brief System Requirements * **Functional Requirements:** The system must allow

users to log in securely based on their roles, enable members to search the catalog, allow librarians to issue/return books, and automatically flag overdue items. * **Non-Functional Requirements:** The system must ensure data consistency, maintain secure user authentication, provide a responsive user interface, and ensure reliable data storage.